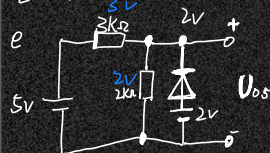
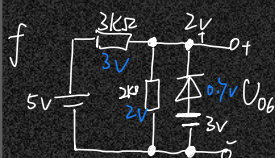


Analog Electronics Assignment

3.4

截止, $U_{o5} = 2V$.

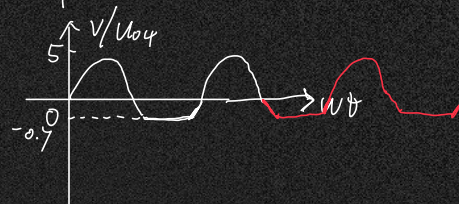
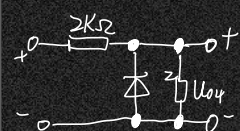
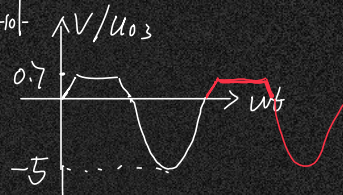
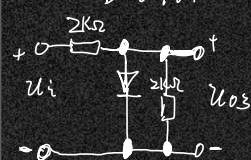
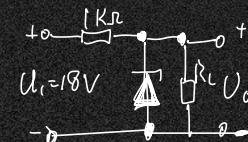
f



导通.

$$U_{o6} - 3 + 0.7 = 0$$

$$U_{o6} = 2.3V$$

3.5. $U_D = 0.7V$. $u_i \rightarrow$ 3.9. $U_Z = 6V$, $I_Z = 5mA$, $I_{ZM} = 30mA$ 

$$18 \times \frac{6}{5}$$

or?

(1) 当 $R_L = 200\Omega$ 时, 稳压电流 $I_{bz} = I_R - I_{RL} = (\frac{18-6}{1} - \frac{6}{0.2}) = -18 \notin (5mA, 30mA)$

$$\therefore I_{bz} \notin [I_Z, I_{ZM}], U_o = \frac{U_1 \cdot 0.2}{(1+0.2)} = 3V$$

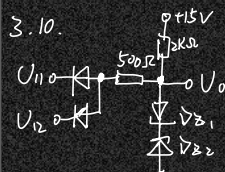
← (假设稳压)

$$\text{当 } R_L = 5k\Omega \text{ 时, } I_{bz} = (\frac{18-6}{1} - \frac{6}{5}) = 10.8mA \Rightarrow U_o = 6V$$

(2) $I_Z \leq I_{bz} \leq I_{ZM} \Rightarrow U_1 \in [12.2, 37.2]$

$$I_{bz} = (\frac{U_1-6}{1} - \frac{6}{5})$$

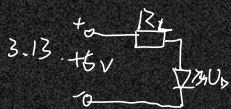
3.10.



U_1/V	U_2/V	D_1	D_2	U_o/V
-15	-15	通	通	-6
-15	+15	通	断	-6
+15	-15	断	通	-6
+15	+15	断	断	+6

Hard to tell, here!

3.13.



$$U_R = U_{in} - U_D = 5 - 1.5 = 3.5V \quad \text{①} \Rightarrow$$

$$5mA \leq \frac{U_R}{R} \leq 20mA \quad \text{②}$$

$$R \in [175\Omega, 700\Omega]$$